PRACTICE SHEET

1. $A=\begin{vmatrix}2a & 3r & x\\ 4b & 6s & 2y\\ -2c & -3t & -z\end{vmatrix}=\lambda\begin{vmatrix}a & r & x\\ b & s & y\\ c & t & z\end{vmatrix}$, then what is the value of λ?

(a) 12 (b) −12
(c) 7 (d) −7

2. What is the value of $\begin{vmatrix}1-i & \omega^2 & -\omega\\ \omega^2+i & \omega & -i\\ 1-2i-\omega^2 & \omega^2-\omega & i-\omega\end{vmatrix}$, where ω is the rube root of unity?

(a) −1 (b) 1
(c) 2 (d) 0

3. If a, b and c are non-zero real numbers and $\begin{vmatrix}1+a & 1 & 1\\ 1 & 1+b & 1\\ 1 & 1 & 1+c\end{vmatrix}=0$, then what is the value of $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}$?

(a) 2 (b) 1
(c) −1 (d) 0

$\begin{vmatrix}y & x & y+z\end{vmatrix}$

$\Delta_1=\begin{vmatrix}a_1 & b_1\\ a_2 & b_2\end{vmatrix},\Delta_2=\begin{vmatrix}b_1 & c_1\\ b_2 & c_2\end{vmatrix},\Delta_3\begin{vmatrix}c_1 & a_1\\ c_2 & a_2\end{vmatrix}$,

Then (x, y) is equal to which one of the following?

(a) $(\Delta_2/\Delta_1, \Delta_3/\Delta_1)$ (b) $(\Delta_3/\Delta_1, \Delta_2/\Delta_1)$
(c) $(\Delta_1/\Delta_2, \Delta_1/\Delta_3)$ (d) $(-\Delta_1/\Delta_2, \Delta_1/\Delta_3)$

9. What is the determinant $\begin{vmatrix}bc & a & a^2\\ ca & b & b^2\\ ab & c & c^2\end{vmatrix}$ equal to?

(a) $\begin{vmatrix}1 & a & a^2\\ 1 & b & b^2\\ 1 & c & c^2\end{vmatrix}$ (b) $\begin{vmatrix}1 & a^2 & a^3\\ 1 & b_2 & b^3\\ 1 & c^2 & c^3\end{vmatrix}$

(c) $\begin{vmatrix}1 & a & a^3\\ 1 & b & b^3\\ 1 & c & c^3\end{vmatrix}$ (d) $\begin{vmatrix}1 & a^2 & a^3\\ 1 & b^2 & b^3\\ 1 & c^2 & c^3\end{vmatrix}$

10. If $x^2+y^2+z^2=1$, then what is the value of $\begin{vmatrix}1 & z & -y\\ -z & 1 & x\\ y & -x & 1\end{vmatrix}$?

(a) 0 (b) 1
(c) 2 (d) 2−2xyz